Neural Network Engine - Technical Documentation

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1. Project Overview

This project implements a fully functional Neural Network engine from scratch using C++.

1.1 Core Philosophy

- **Zero Dependency**: Standard C++ libraries only.
- **Manual Memory**: Move Semantics (O(1)).
- **Efficiency**: Optimized Matrix operations.

2. Architecture

2.1 Matrix Engine

Custom linear algebra engine with move constructors.

2.2 Layer Class

Managed weights, biases, and ReLU activation.

3. Mathematical Foundations

Forward Pass:

$$Z = W \cdot X + B$$

$$A = \mathrm{ReLU}(Z)$$

Backward Pass (Gradient Descent):

$$\nabla W = \text{error} \cdot X^T$$

4. Current Status

- **XOR Problem**: Solved.
- **Automation**: PDF auto-generated.

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